

Question-5

- Below are the commands I used to obtain the firewall:-
- I have assumed that database ports are 3306 for MySQL, 27017 for MongoDB, and 5432 for PostgreSQL.
- For the internal IP ranges, I have assumed them from 191.168.1.0/24, though to allow all private IP addresses, it should be 191.168.0.0/16.

First, I dropped the packets for forwarding, ensuring that unauthorized traffic doesn't pass through the server.

Next, I ensured that already established connections were accepted.

```
sudo iptables -A FORWARD -j DROP
```

```
sudo iptables -A INPUT -m conntrack --ctstate ESTABLISHED -j ACCEPT
```

Next, I ensured that admin IP is able to access the SSH (22) and RDT (3389)

```
sudo iptables -A INPUT -p tcp -s 192.168.1.3 --dport 22 -j ACCEPT
```

```
sudo iptables -A INPUT -p tcp -s 192.168.1.3 --dport 3389 -j ACCEPT
```

Next, I blocked all the other traffic trying to access SSH and RDT and logged these too.

```
sudo iptables -A INPUT -p tcp --dport 3389 -j LOG --log-prefix "Unauthorised Access to RDP!"
```

```
sudo iptables -A INPUT -p tcp --dport 3389 -j DROP
```

```
sudo iptables -A INPUT -p tcp --dport 22 -j LOG --log-prefix "Unauthorised Access to SSH!"
```

```
sudo iptables -A INPUT -p tcp --dport 22 -j DROP
```

Next, I blocked the blacklisted IPs to access HTTP and HTTPS.

```
sudo iptables -A INPUT -p tcp -s 103.25.231.0/24 --dport 80 -j DROP
```

```
sudo iptables -A INPUT -p tcp -s 103.25.231.0/24 --dport 443 -j DROP
```

Next, I rate-limited five connections per IP for HTTP and HTTPS, this has to be done before allowing genuine access to HTTP and HTTPS, as these genuine access may have crossed the limits too.

```
sudo iptables -A INPUT -p tcp --dport 80 -m state --state NEW -m hashlimit --hashlimit-above 5/sec --hashlimit-mode srcip --hashlimit-name rate_limit_per_ip -j DROP
```

```
sudo iptables -A INPUT -p tcp --dport 443 -m state --state NEW -m hashlimit --hashlimit-above 5/sec --hashlimit-mode srcip --hashlimit-name rate_limit_per_ip_https -j DROP
```

Now, I allowed the traffic for HTTPS and HTTP.

```
sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
```

```
sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT
```

Next, I allowed the private IPs to access the database services.

```
sudo iptables -A INPUT -p tcp -s 192.168.1.0/24 --dport 3306 -j ACCEPT
sudo iptables -A INPUT -p tcp -s 192.168.1.0/24 --dport 27017 -j ACCEPT
sudo iptables -A INPUT -p tcp -s 192.168.1.0/24 --dport 5432 -j ACCEPT
```

Now, after allowing the genuine IPs to access the database, I blocked all the other IPs and logged this activity, too.

```
sudo iptables -A INPUT -p tcp --dport 3306 -j LOG --log-prefix "Unauthorised Access to SQL!"
sudo iptables -A INPUT -p tcp --dport 27017 -j LOG --log-prefix "Unauthorised Access to MONGODB!"
sudo iptables -A INPUT -p tcp --dport 5432 -j LOG --log-prefix "Unauthorised Access to POSTGRESQL!"
sudo iptables -A INPUT -p tcp --dport 3306 -j DROP
sudo iptables -A INPUT -p tcp --dport 27017 -j DROP
sudo iptables -A INPUT -p tcp --dport 5432 -j DROP
```

After allowing genuine cases for everything, I now blocked all the traffic and logged this, too.

```
sudo iptables -A INPUT -j LOG --log-prefix "Access Denied & Packets Dropped"
sudo iptables -A INPUT -j DROP
```

- Below is the ss for the same

```
nikhil@nikhil-linux:~$ sudo iptables -A FORWARD -j DROP
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -m conntrack --ctstate ESTABLISHED -j ACCEPT
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp -s 192.168.1.3 -dport 22 -j ACCEPT
Bad argument '22'
Try 'iptables -h' or 'iptables --help' for more information.
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp -s 192.168.1.3 --dport 22 -j ACCEPT
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp -s 192.168.1.3 --dport 3389 -j ACCEPT
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp --dport 3389 -j LOG --log-prefix "Unauthorised Access to RDP!"
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp --dport 3389 -j DROP
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp --dport 22 -j LOG --log-prefix "Unauthorised Access to SSH!"
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp --dport 22 -j DROP
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp -s 103.25.231.0/24 --dport 80 -j DROP
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp -s 103.25.231.0/24 --dport 443 -j DROP
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp --dport 80 -m state --state NEW -m hashlimit --hashlimit-above 5/sec --hashlimit-mode srcip --hashlimit-name rate_limit_per_ip -j DROP
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp --dport 443 -m state --state NEW -m hashlimit --hashlimit-above 5/sec --hashlimit-mode srcip --hashlimit-name rate_limit_per_ip_https -j DROP
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp -s 192.168.1.0/24 --dport 3306 -j ACCEPT
iptables v1.8.10 (nf_tables): no command specified
Try 'iptables -h' or 'iptables --help' for more information.
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp -s 192.168.1.0/24 --dport 3306 -j ACCEPT
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp -s 192.168.1.0/24 --dport 27017 -j ACCEPT
nikhil@nikhil-linux:~$
```

```

nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp --dport 3389 -j DROP
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp --dport 22 -j LOG --log-prefix "Unauthorised Access to SSH!"
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp --dport 22 -j DROP
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp -s 103.25.231.0/24 --dport 80 -j DROP
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp -s 103.25.231.0/24 --dport 443 -j DROP
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp --dport 80 -m state --state NEW -m hashlimit --hashlimit-above 5/sec --hashlimit-mode srcip --hashlimit-name rate_limit_per_ip -j DROP
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp --dport 443 -m state --state NEW -m hashlimit --hashlimit-above 5/sec --hashlimit-mode srcip --hashlimit-name rate_limit_per_ip_https -j DROP
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT
nikhil@nikhil-linux:~$ sudo iptables -p tcp -s 192.168.1.0/24 --dport 3306 -j ACCEPT
iptables v1.8.10 (nf_tables): no command specified
Try 'iptables -h' or 'iptables --help' for more information.
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp -s 192.168.1.0/24 --dport 3306 -j ACCEPT
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp -s 192.168.1.0/24 --dport 27017 -j ACCEPT
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp -s 192.168.1.0/24 --dport 5432 -j ACCEPT
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp --dport 3306 -j LOG --log-prefix "Unauthorised Access to SQL!"
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp --dport 5432 -j LOG --log-prefix "Unauthorised Access to MONGODB!"
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp --dport 5432 -j LOG --log-prefix "Unauthorised Access to PostgreSQL!"
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp --dport 3306 -j DROP
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp --dport 27017 -j DROP
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -p tcp --dport 5432 -j DROP
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -j LOG --log-prefix "Access Denied & Packets Dropped"
nikhil@nikhil-linux:~$ sudo iptables -A INPUT -j DROP
nikhil@nikhil-linux:~$

```

- Following is the final iptables with line numbers:-

❑ INPUT CHAIN

```

nikhil@nikhil-linux:~$ sudo iptables -L INPUT --line-numbers
Chain INPUT (policy ACCEPT)
num target prot opt source destination
1 ACCEPT all -- anywhere anywhere ctstate ESTABLISHED
2 ACCEPT tcp -- 192.168.1.3 anywhere tcp dpt:ssh
3 ACCEPT tcp -- 192.168.1.3 anywhere tcp dpt:ms-wbt-server
4 LOG tcp -- anywhere anywhere tcp dpt:ms-wbt-server LOG level warn prefix "Unauthorised Access to RDP!"
5 DROP tcp -- anywhere anywhere tcp dpt:ms-wbt-server
6 LOG tcp -- anywhere anywhere tcp dpt:ssh LOG level warn prefix "Unauthorised Access to SSH!"
7 DROP tcp -- anywhere anywhere tcp dpt:ssh
8 DROP tcp -- 103.25.231.0/24 anywhere tcp dpt:http
9 DROP tcp -- 103.25.231.0/24 anywhere tcp dpt:https
10 DROP tcp -- anywhere anywhere tcp dpt:http state NEW limit: above 5/sec burst 5 mode srcip
11 DROP tcp -- anywhere anywhere tcp dpt:https state NEW limit: above 5/sec burst 5 mode srcip
12 ACCEPT tcp -- anywhere anywhere tcp dpt:http
13 ACCEPT tcp -- anywhere anywhere tcp dpt:https
14 ACCEPT tcp -- 192.168.1.0/24 anywhere tcp dpt:mysql
15 ACCEPT tcp -- 192.168.1.0/24 anywhere tcp dpt:27017
16 ACCEPT tcp -- 192.168.1.0/24 anywhere tcp dpt:postgresql
17 LOG tcp -- anywhere anywhere tcp dpt:mysql LOG level warn prefix "Unauthorised Access to SQL!"
18 LOG tcp -- anywhere anywhere tcp dpt:27017 LOG level warn prefix "Unauthorised Access to MONGODB"
19 LOG tcp -- anywhere anywhere tcp dpt:postgresql LOG level warn prefix "Unauthorised Access to PostgreSQL"

```

```
nikhil@nikhil-linux: ~  
ess to RDP!"  
5 DROP tcp -- anywhere anywhere tcp dpt:ms-wbt-server  
6 LOG tcp -- anywhere anywhere tcp dpt:ssh LOG level warn prefix "Unauthorised Access to SSH  
!"  
7 DROP tcp -- anywhere anywhere tcp dpt:ssh  
8 DROP tcp -- 103.25.231.0/24 anywhere tcp dpt:http  
9 DROP tcp -- 103.25.231.0/24 anywhere tcp dpt:https  
10 DROP tcp -- anywhere anywhere tcp dpt:http state NEW limit: above 5/sec burst 5 mode srcip  
11 DROP tcp -- anywhere anywhere tcp dpt:https state NEW limit: above 5/sec burst 5 mode srcip  
12 ACCEPT tcp -- anywhere anywhere tcp dpt:http  
13 ACCEPT tcp -- anywhere anywhere tcp dpt:https  
14 ACCEPT tcp -- 192.168.1.0/24 anywhere tcp dpt:mysql  
15 ACCEPT tcp -- 192.168.1.0/24 anywhere tcp dpt:27017  
16 ACCEPT tcp -- 192.168.1.0/24 anywhere tcp dpt:postgresql  
17 LOG tcp -- anywhere anywhere tcp dpt:mysql LOG level warn prefix "Unauthorised Access to S  
QL!"  
18 LOG tcp -- anywhere anywhere tcp dpt:27017 LOG level warn prefix "Unauthorised Access to M  
ONGOD"  
19 LOG tcp -- anywhere anywhere tcp dpt:postgresql LOG level warn prefix "Unauthorised Access  
to POSTGR"  
20 DROP tcp -- anywhere anywhere tcp dpt:mysql  
21 DROP tcp -- anywhere anywhere tcp dpt:27017  
22 DROP tcp -- anywhere anywhere tcp dpt:postgresql  
23 LOG all -- anywhere anywhere LOG level warn prefix "Access Denied & Packets Dropp"  
24 DROP all -- anywhere anywhere  
nikhil@nikhil-linux:~$
```

❑ FORWARD CHAIN

```
nikhil@nikhil-linux: ~  
nikhil@nikhil-linux:~$ sudo iptables -L FORWARD --line-numbers  
Chain FORWARD (policy ACCEPT)  
num target prot opt source destination  
1 DROP all -- anywhere anywhere  
nikhil@nikhil-linux:~$
```

References

- Linux Man Page for iptables: <https://linux.die.net/man/8/iptables>
- Hashlimit Module for iptables:
<https://poorlydocumented.com/2017/08/understanding-iptables-hashlimit-module/>